Dynamic games

Dynamic games have multiple time-steps (rather than a single decision leading to a payoff, as in the Prisoner's Dilemma for example). An example is **Linear Quadratic (LQ) Games** which are a class of dynamic games involving multiple decision-makers (players) where each player aims to minimize a quadratic cost function subject to linear dynamics. They extend the Linear Quadratic Regulator (LQR) framework to game-theoretic settings.

State-space:

$$x_{k+1} = Ax_k + \sum_{i=1}^{N} B_i u_k^i$$

where x_k is the state at time *k*, A, B_i are system matrices and u_k^i is the control input (decision) for player *i* at time *k*.

Each player is trying to minimise their own cost:

$$J^i = \sum_{k=0}^{\infty} \left(x_k^\top Q^i x_k + u_k^{i\top} R^i u_k^i \right)$$

Via a feedback law

$$u_k^i = -K^i x_k$$

Similarly to standard LQR control, there is an Algebraic Riccati Equation which can be solved to find *K* such that *u* is the Nash equilibrium of the dynamic game.

Finance applications

Stackelberg game

Suppose two firms are producing the same product and competing. The market price depends on the quantity produced, as follows:

$$P(Q) = a - bQ, \text{where } Q = q_1 + q_2$$

Where *P* is the market price, *a, b* are constants, and *Q* is the quantity produced by both firms 1 (leader) and 2 (follower). How much should each firm produce?

Profit (= - cost) function:

$$\pi_1 = (P - c)q_1 = (a - b(q_1 + q_2) - c)q_1$$

$$\pi_2 = (P - c)q_2 = (a - b(q_1 + q_2) - c)q_2$$

In a Stackelberg game, the leader chooses their decision first and the follower adopts the **best** response. Often, there is a first-mover advantage.

The follower's response will be optimised based on the leader's decision (q_1).

$$\frac{d\pi_2}{dq_2} = a - b(q_1 + 2q_2) - c = 0$$

$$q_2 = \frac{a - c - bq_1}{2b}$$

The leader makes their move assuming an optimal response by the follower:

$$\pi_1 = \left(a - b\left(q_1 + \frac{a - c - bq_1}{2b}\right) - c\right) q_1$$

The optimal quantities are:

$$q_1^* = \frac{a - c}{2b}, \qquad q_2^* = \frac{a - c}{4b}$$

This may be compared to the other major economic game-theoretic model, the Cournot model, which assumes simultaneous decisions, resulting in:

$$q_1^* = \frac{a - c}{3b}, \qquad q_2^* = \frac{a - c}{3b}$$